Ryan Brundage
Queue Analysis for Mickey's Department Store
Date: 04/07/26

Abstract:

This report analyzes Mickey's Department Stores' checkout system during peak holiday periods, including theoretical queueing models and calculations as well as simulation results. The system is modeled as an M/M/s (or k) queue with one line that feeds into multiple server stations. The objective of this is to determine the minimum number of registers required to meet the requirement that no more than 10% of customers wait longer than 5 minutes to be served. The results show that 7 registers meet this requirement under the baseline conditions. Howeveerr when service times increase, the store requires more registers than the building can support.

Introduction:

Mickey's Department Store experiences a high influx of customers during peak holiday times, which has led to long checkout lines and customers being dissatisfied in the past. The store implemented a single-line queue that feeds multiple registers to help combat this. The goal of the store's management is to determine how many registers are required to support the high customer traffic and maintain service levels. From past data, customers arrive about once every 75 seconds, with an average service time of about 6.5 minutes per customer. The busy time lasts for about 4 hours. Management requires that no more than 10% of customers wait for longer than 5 minutes before reaching a server. The store's layout allows for up to 8 registers.

Baseline Assumptions and Model:

The checkout system is modeled as an M/M/s (s used in place of k to denote "serving stations"), with arrivals following a Poisson process and service times following an exponential distribution. The arrival rate, λ, is 0.8 customers per minute, found by converting 1 every 75 seconds to minutes. The service rate, μ, is the number of customers a single server can process per minute, or 1 divided by 6.5 minutes. The system has s identical servers where s is the number of registers operating.

The utilization of the system, ρ, is the ratio of the arrival rate to the total service capacity and indicates how busy the system is. This is seen in equation 1: $\rho = \lambda / s * \mu$ (1)
Where λ is the arrival rate, μ is the service rate per register, and s is the number of registers. A system is stable if utilization is less than 1; otherwise, it is overloaded.

The probability that there are zero customers in the system, P_0, is given by equation 2:

$P_0 = [\sum_{n=0}^{s-1} \frac{\left(\frac{\lambda}{\mu}\right)^n}{n!} + \frac{\left(\frac{\lambda}{\mu}\right)^s}{s!} * \frac{1}{1-\frac{\lambda}{s\mu}}]$ (2) where P_0 represents the steady-state probability of

having no customers in the system. In the equation, n represents the number of customers in the system.

The expected number of customers in the queue L_q is: $L_q = \frac{(\frac{\lambda}{\mu})^s \rho}{s!(1-\rho)} P_0$ (3) The expected number of total customers in the system L is: $L = L_q + \frac{\lambda}{\mu}$ (4), which includes both customers in the queue and those being served.

The expected waiting time in the queue is given by: $W_q = \frac{L_q}{\lambda}$ (5) . The total expected time in the system is denoted by W and derived from: $W = \frac{L}{\lambda}$ (6)

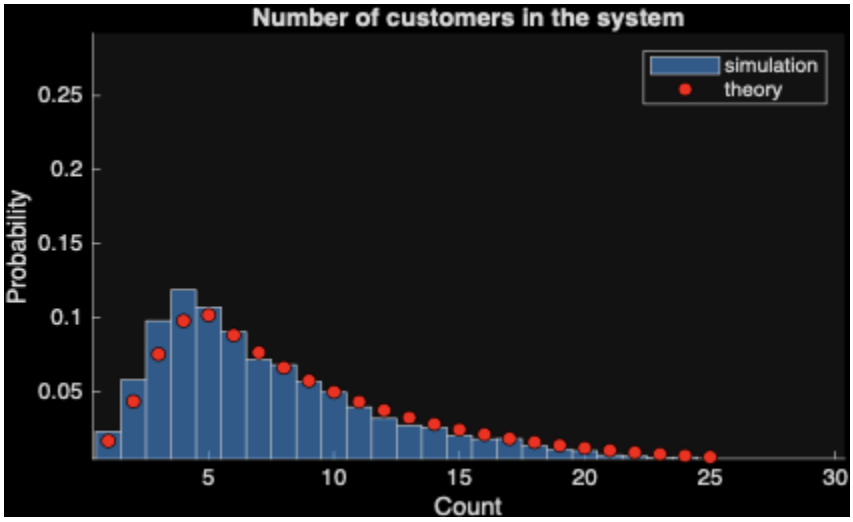
Alongside these standard measures, the probability that a customer waits more than 5 minutes is computed using a simulation model and theoretical approximation in MATLAB. The s values also start at s = 6 since the offered load $\frac{\lambda}{\mu} = 5.2$.

Baseline Results:

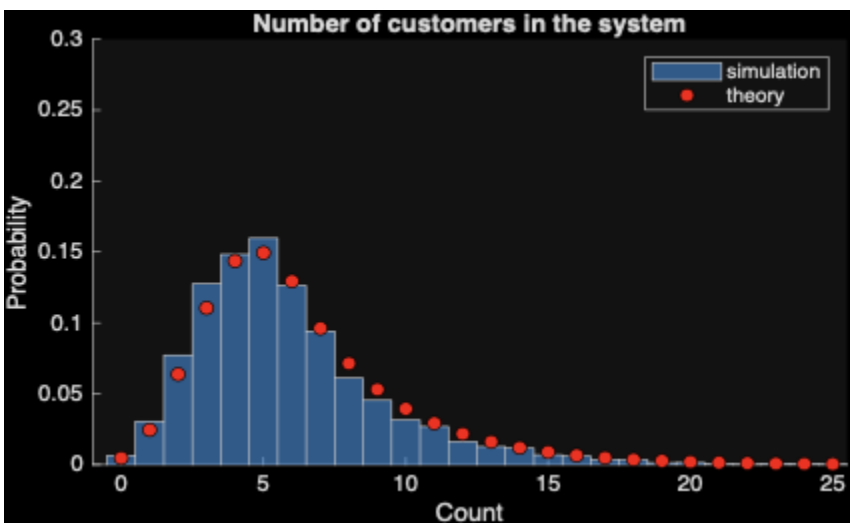
The system was evaluated at different values of s. The s values also start at s = 6 since the offered load $\frac{\lambda}{\mu} = 5.2$. When s=6, the system operates at very high utilization $\rho = 0.8667$ and experiences significant congestion. The theoretical probability that a customer waits more than 5 minutes is 0.3576. The simulation result for this probability is 0.2779, both well above the required threshold of 0.10. The difference between them is -0.0796. The expected queue length is $L_q = 8.0752$. The expected waiting time in the queue is $W_q = 3.5238$. The totals W and L are W = 10.0238 and L = 8.0752.

When s=7, the system operates at $\rho = 0.7429$ a moderate rate with slight congestion. The theoretical probability that a customer waits more than 5 minutes is 0.0937. The simulation result for this probability is 0.0700, both just under the required threshold of 0.10. The difference between them is -0.0236. The expected queue length is $L_q = 5.8221$. The expected waiting time in the queue is $W_q = 0.7738$. The totals W and L are W = 7.2738 and L = 5.8221. Having 7 serving stations is the minimum required to meet the requirements, and there is still enough room since the register capacity is 8.

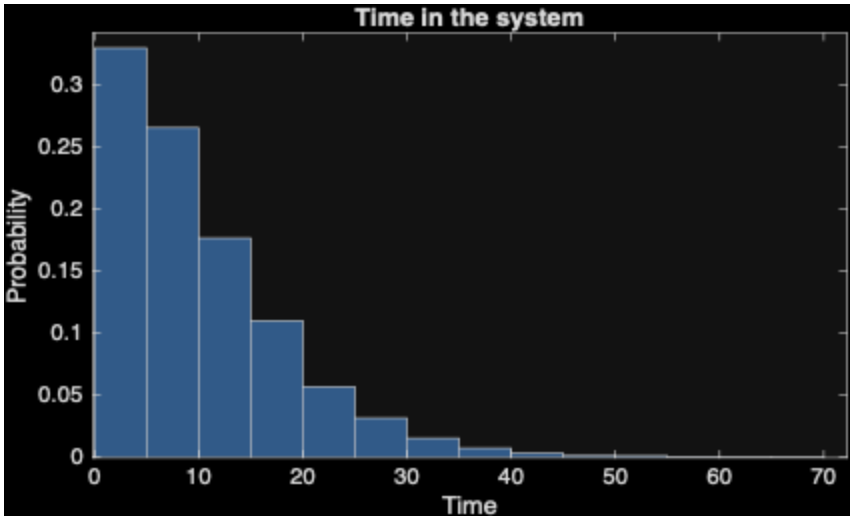
Number of Customers in The System s=6:



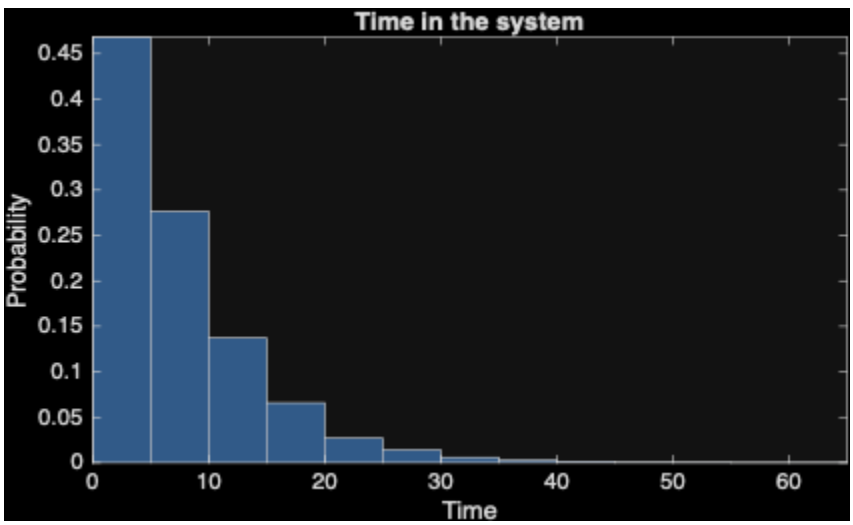
Number of Customers in System s=7



Time in System s=6:



Time In System s=7:



Sensitivity Analysis:

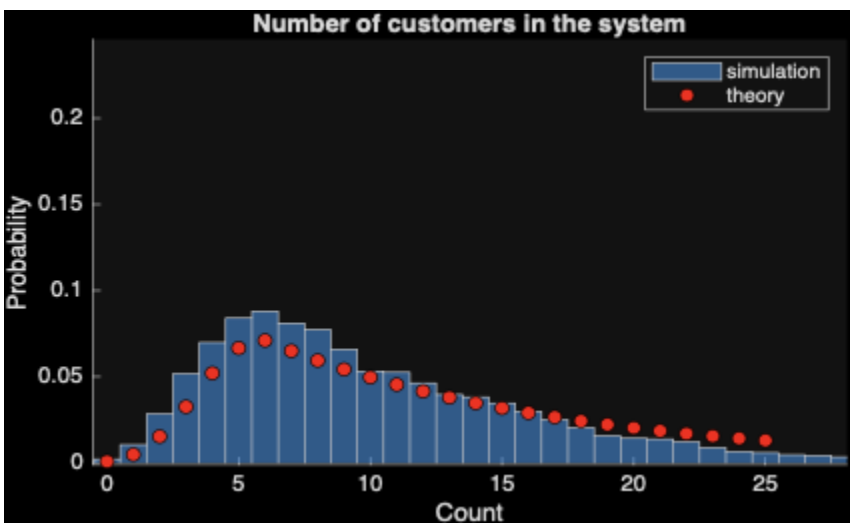
When the average service time increases from 6.5 minutes to 8 minutes, the service rate decreases and utilization increases, resulting in greater congestion. We start at s = 7 since the offered load $\frac{\lambda}{\mu} = 6.4$.

When s=7, the system operates at very high utilization $\rho = 0.9143$ and experiences significant congestion. The theoretical probability that a customer waits more than 5 minutes is 0.5204. The simulation result for this probability is 0.3453, both well above the required threshold of 0.10. The difference between them is -0.1751. The expected queue length is $L_q = 3.5585$. The expected waiting time in the queue is $W_q = 4.4482$. The totals W and L are W = 12.4482 and L = 10.1465.

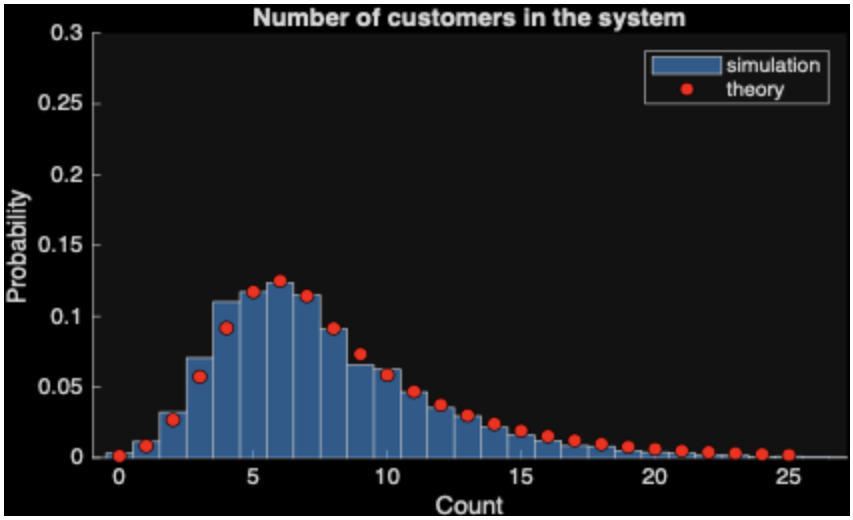
When s=8, the system operates at very high utilization $\rho = 0.8000$ and experiences significant congestion. The theoretical probability that a customer waits more than 5 minutes is 0.1684. The simulation result for this probability is 0.1208, which is still above the required threshold of 0.10. The difference between them is -0.0476. The expected queue length is $L_q = 1.0931$. The expected waiting time in the queue is $W_q = 1.3664$. The totals W and L are W = 9.3664 and L = 7.5846.

When s=9, the system operates at $\rho = 0.7111$ a moderate rate with slight congestion. The theoretical probability that a customer waits more than 5 minutes is 0.0516. The simulation result for this probability is 0.0342, which is below the required threshold of 0.10. The difference between them is -0.0174. The expected queue length is $L_q = 0.2325$. The expected waiting time in the queue is $W_q = 0.2906$. The totals W and L are W = 8.2906 and L = 6.6898. Having 9 serving stations is the minimum required to meet the requirements; there is not enough space in the layout to fit that many registers.

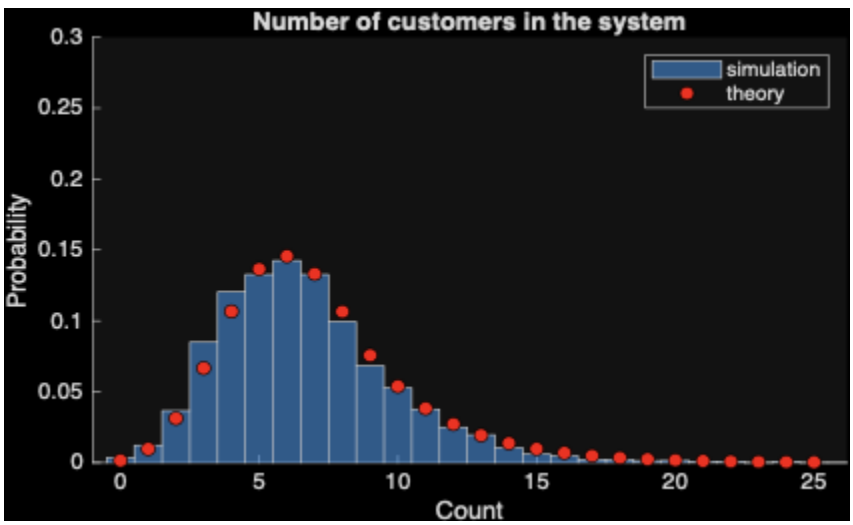
Number in System s=7, 8 Minute Service Time:



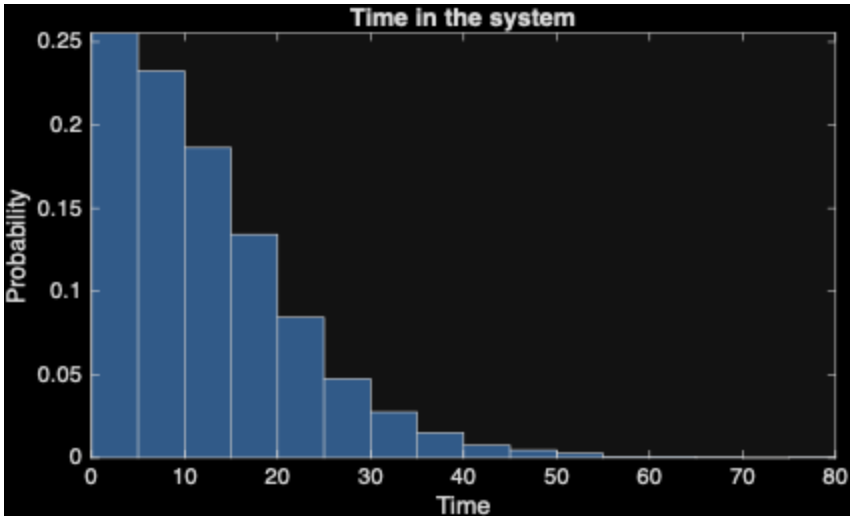
Number in System s=8, 8 Minute Service Time:



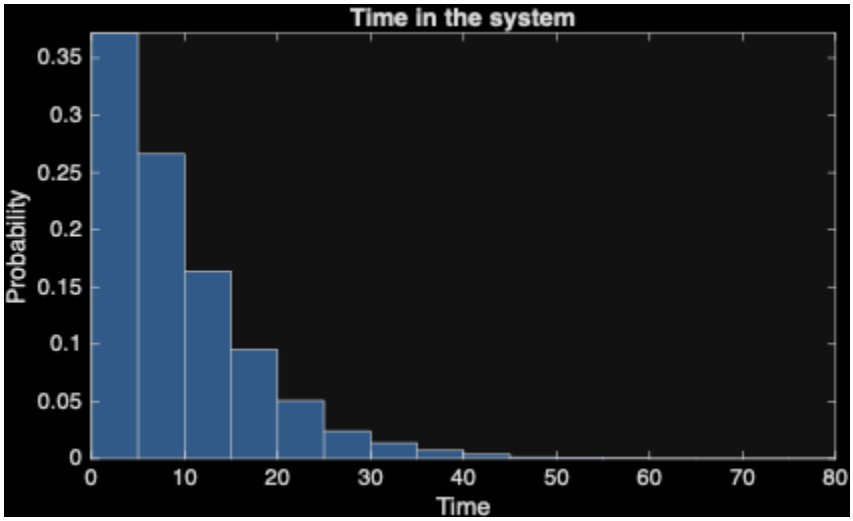
Number in System s=9, 8 Minute Service Time:



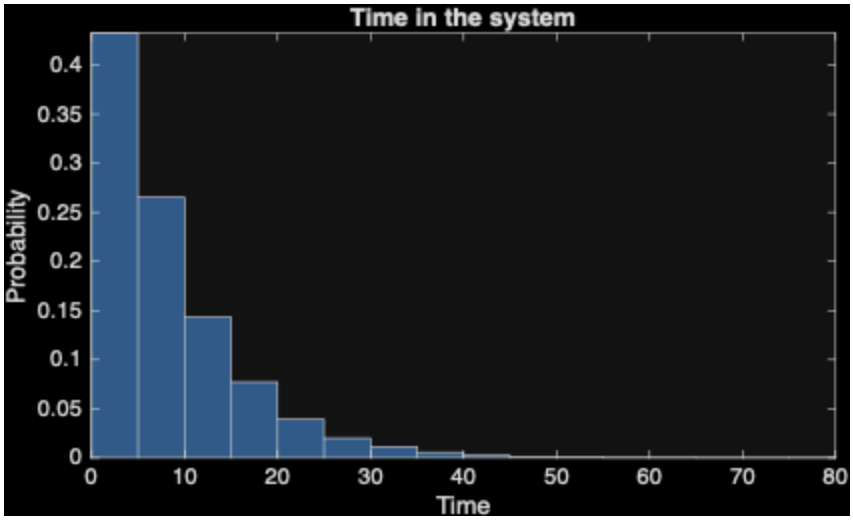
Time in System s=7, 8 Minute Service Time



Time in System s=8, 8 Minute Service Time



Time in System s=9, 8 Minute Service Time



Recommendations:

The analysis shows that Mickeys should operate with at least s=7 registers during peak holiday hours if the service time remains near 6.5 minutes per customer. Both the theoretical and simulation results confirm that this number of registers satisfies the 10% requirement. If the service time increases to 8 minutes per customer, neither the theoretical nor the simulation results meet the 10% requirement, given the available space in the store. In this case, management should focus on improving employee efficiency rather than increasing the number of registers. I am confident in my results since the theoretical and simulation differences are not too large. I say 'moderately' because there is always room for variation in real-world scenarios, and since the simulation runs for only 4 hours, it could be more accurate.